NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

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Master in Computer Science and Engineering

COST AND OPERATIONS OPTIMIZATION IN MULTI-CLOUD ENVIRONMENTS OVER GENAI FRAMEWORKS

MASTER IN COMPUTER SCIENCE AND ENGINEERING

SPECIALIZATION IN SPECIALIZATION NAME

NOVA University Lisbon

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Abstract

Keywords: Multi-Cloud · Generative AI · Cost Optimisation · Cloud Operations · GenAI Frameworks

Resumo

Palavras-chave: Multi-Cloud · Inteligência Artificial Generativa · Otimização de Custos · Operações em Nuvem · Frameworks GenAI

Disclosure of Delegation to Generative Artificial Intelligence

Duarte José Lourenço Belo declares[1] the use of Generative AI in the research work and writing process of this document. According to the GAIDeT taxonomy [7], the following tasks were delegated to Generative AI tools under full human supervision:

Conceptualization

- [x] Idea generation
- [x] Defining the research objective
- [x] Formulating research questions and hypotheses
- [ ] Feasibility assessment and risk evaluation
- [ ] Preliminary hypothesis testing

Literature Review

- [x] Literature search and systematization
- [x] Writing the literature review
- [ ] Analysis of market trends and/or patent environment
- [ ] Evaluation of novelty and identification of gaps

Methodology

- [x] Research design
- [x] Development of experimental or research protocols
- [ ] Selection of research methods

Software Development and Automation

- [x] Code generation
- [x] Code optimization
- [ ] Process automation
- [ ] Creation of algorithms for data analysis

Data Management

- [x] Data collection
- [x] Validation
- [x] Data cleaning
- [ ] Data curation and organization
- [ ] Data analysis
- [ ] Visualization
- [ ] Reproducibility testing

[1] This declaration was generated using LuaLaTeX [8] and the LaTeX package `aidisclose` [4].

Writing and Editing

- [x] Text generation
- [x] Proofreading and editing
- [x] Summarizing text
- [ ] Formulation of conclusions
- [ ] Adapting and adjusting emotional tone
- [ ] Translation
- [ ] Reformatting
- [ ] Press releases and outreach materials

Ethics Review

- [x] Bias analysis and discrimination assessment
- [x] Ethical risk analysis
- [x] Monitoring compliance with ethical standards
- [ ] Data confidentiality monitoring

Supervision

- [ ] Quality assessment
- [ ] Trend identification
- [ ] Identification of limitations
- [ ] Recommendations
- [ ] Publication support

Generative AI tools used: ChatGPT-5.2, Gemini 3 Pro, Claude 4.5 Opus, and Perplexity 12.12.25.

Additional comment #1: The AI was used primarily for refining the code in Section 3, but the algorithm logic was derived manually.

Additional comment #2: This is a second additional comment!

Additional comment: This is a third additional comment! It is starred and, thus, does not get a number. Also, this comment is longer and, hopefully, will span to the next line!

Additional comment #3: This is the third non-starred additional comment!

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Background

2.1 Foundations of Neural Language Representation

The ability to represent language in a form that machines can process and reason over is a prerequisite for any modern Generative AI system. Two landmark contributions established the theoretical and practical foundations upon which contemporary large language models (LLMs) and retrieval-augmented architectures are built: the introduction of dense word representations via neural networks, and the development of the Transformer architecture based solely on attention mechanisms.

2.1.1 Distributed Word Representations

Prior to 2013, natural language processing relied predominantly on sparse, high-dimensional representations such as one-hot encodings, which offered no intrinsic notion of semantic similarity between words. Mikolov et al. [5] addressed this limitation by proposing Word2Vec, a computationally efficient framework for learning continuous vector representations of words from large unlabelled corpora. The model is trained through a self-supervised prediction task using one of two architectures: Continuous Bag-of-Words (CBOW), which predicts a target word from its surrounding context, or Skip-gram, which performs the inverse prediction. The resulting word embeddings encode both syntactic and semantic structure in a dense vector space, enabling linear algebraic operations over meaning — most notably the well-known relationship $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$.

The significance of Word2Vec extends well beyond the paper itself. By demonstrating that semantic similarity can be captured as geometric proximity in a continuous vector space, it established the distributional hypothesis as an actionable engineering principle. This insight directly underlies the design of modern embedding models and vector databases, which are central components in Retrieval-Augmented Generation (RAG) pipelines [3]. In the context of this work, the quality and cost of the embedding stage in the Maestro framework's RAG pipeline is a direct consequence of choices made in this representational paradigm.

2.1.2 The Transformer Architecture

Despite the advances in distributed representations, sequence modelling remained constrained by recurrent architectures such as LSTMs and GRUs, which process tokens sequentially and suffer from vanishing gradients over long contexts. Vaswani et al. [10] proposed a radical departure: the Transformer, an architecture that dispenses with recurrence entirely and relies solely on attention mechanisms to model dependencies between all positions in a sequence simultaneously. The core operation, scaled dot-product attention, computes output representations as a weighted sum of value vectors, where weights are derived from the compatibility between query and key projections:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

By running multiple such attention operations in parallel — Multi-Head Attention — the model learns to simultaneously attend to different types of relationships across the input, such as syntactic dependencies and semantic co-references. Positional encodings are injected additively into the input embeddings to compensate for the architecture's inherent permutation invariance. The resulting model achieved 28.4 BLEU on the WMT 2014 English-to-German translation benchmark, surpassing all prior approaches while requiring a fraction of the training compute [10].

The Transformer's parallelism over sequence positions unlocked a new scaling regime for neural language models. Architectures derived from it — most notably encoder-only models such as BERT [2] for representation learning, and decoder-only models such as the GPT series [1] for text generation — became the standard building blocks of enterprise GenAI systems. Critically, the attention mechanism scales quadratically with sequence length, $O(n^2)$, meaning that context length is a primary cost driver in token-priced LLM APIs. This architectural property is foundational to the FinOps strategies explored in this dissertation, namely prompt compression, semantic caching, and dynamic model selection, as each seeks to reduce the effective n processed by the model or the frequency with which inference is invoked at all [5, 10].

Together, these two works define the representational substrate of modern GenAI: Word2Vec established that meaning can be encoded as position in a learned vector space, and the Transformer established that context-sensitive reasoning over such representations can be performed efficiently and at scale. The remainder of this chapter examines the architectural patterns, operational frameworks, and cost optimisation techniques that build upon these foundations in production-grade enterprise systems.

2.2 Large Language Models in Production

The preceding section established the two foundational breakthroughs that made modern language AI possible: distributed word representations and the Transformer architecture. Building upon these primitives, the field witnessed a rapid consolidation around a

unified paradigm — the large language model (LLM) — in which ever-larger Transformer-based networks, pre-trained on internet-scale corpora, acquire general-purpose language competence that can be redirected toward downstream tasks with minimal additional supervision. This section characterizes LLMs as a class, traces the methodological evolution from raw pre-training to instruction-following alignment, and examines the architectural and economic implications of deploying such systems at enterprise scale.

2.2.1 Large Language Models: An Overview

A large language model is a neural language model of sufficient parameter count and training data scale that general linguistic and reasoning capabilities emerge as a consequence of the pre-training objective alone, without task-specific architectural modification [1]. Pre-training proceeds via self-supervised next-token prediction over corpora of hundreds of billions to trillions of tokens, optimizing the model to assign high probability to naturally occurring text. The resulting parameter weights encode an implicit representation of syntax, semantics, world knowledge, and rudimentary reasoning chains that can be elicited through natural language prompting.

LLMs are predominantly realized as *decoder-only* Transformer architectures [10], in which the self-attention layers apply a causal mask that restricts each token's attention to its leftward context, preserving the autoregressive generation property. This design choice contrasts with the encoder-only variant typified by BERT [2], which produces bidirectional contextual representations optimized for discriminative tasks. For generative applications — intelligent assistants, automated reasoning, and document synthesis — the decoder-only paradigm dominates, and it is this class of model that underpins the enterprise GenAI systems examined in this dissertation.

A defining empirical finding in the development of LLMs is the existence of *scaling laws*: systematic, power-law relationships between model size, training compute, dataset volume, and downstream performance [1]. Scaling laws imply that performance is largely predictable and improvable by increasing any of the three axes, establishing a framework within which investment in compute translates into measurable capability gains. This property has driven the successive scaling of commercial models and, critically, has also made token consumption a dominant cost variable: larger models process the same input more expensively, directly motivating the cost optimization strategies central to this work.

2.2.2 From Pre-training to Instruction Tuning

Despite the generative competence acquired during pre-training, early LLMs exhibited a systematic misalignment between model behavior and user intent. A model trained solely to predict the next token on a web-scale corpus learns to continue text in statistically plausible ways, but not necessarily in ways that are *helpful*, *honest*, or *harmless* when deployed as an interactive assistant. Ouyang et al. [6] addressed this gap by introducing a three-stage fine-tuning pipeline centered on Reinforcement Learning from Human Feedback (RLHF).

The procedure operates as follows. First, a *supervised fine-tuning* (SFT) phase trains the base model on high-quality demonstrations of desired behavior, produced by human labelers on a curated prompt distribution. Second, a *reward model* is trained to predict human preference scores over pairs of model outputs, internalizing a proxy for human judgment. Third, the SFT model is further optimized against the reward model signal using Proximal Policy Optimization (PPO), an online reinforcement learning algorithm. The resulting InstructGPT models, despite comprising as few as 1.3 billion parameters, were consistently preferred by human evaluators over the unaligned 175-billion parameter GPT-3 baseline, demonstrating that alignment quality is not strictly a function of scale [6]. This result has profound practical implications: it established RLHF as the standard post-training procedure for production LLMs and confirmed that an aligned smaller model can match or exceed the utility of a larger unaligned one — a relationship that informs the dynamic model selection strategies explored in this dissertation.

2.2.3 Open-Weight Models and the Deployment Spectrum

The commercial LLM landscape is stratified between proprietary, API-accessed models and openly released, self-hostable alternatives. The former class, exemplified by OpenAI's GPT-4 family and Anthropic's Claude series, exposes inference through token-priced cloud APIs; the latter is represented most prominently by Meta AI's Llama series. Touvron et al. [9] introduced Llama 2, a family of models ranging from 7 to 70 billion parameters, pre-trained on two trillion tokens and released under a license permitting commercial use. Fine-tuned variants — Llama 2-Chat — further apply RLHF alignment, achieving performance competitive with closed-source alternatives on standard benchmarks while remaining locally deployable [9].

The existence of capable open-weight models introduces a consequential degree of freedom into enterprise GenAI architectures. Rather than committing uniformly to a single proprietary provider, organizations may route inference dynamically across a portfolio of models differentiated by capability tier, latency characteristics, and per-token cost. Smaller open-weight models, deployed on-premise or on reserved cloud instances, can serve high-volume, lower-complexity queries at substantially reduced marginal cost, while API-accessed frontier models handle tasks requiring peak reasoning performance. This model routing paradigm — often referred to as *LLM cascading* or *dynamic model selection* — is a central cost optimization mechanism evaluated in this dissertation in the context of the Maestro framework's multi-cloud deployment [9, 6].

Together, the progression from raw pre-training through RLHF alignment to the open-weight deployment spectrum defines the operational environment within which enterprise GenAI frameworks must function. The ability to select, combine, and switch between models of varying capability and cost is not merely an architectural convenience but a financial imperative in production systems where token consumption can scale three to five times per client without deliberate optimization [6].

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Bibliography

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. "Language Models are Few-Shot Learners". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. 2020, pp. 1877–1901. URL: https://arxiv.org/abs/2005.14165 (cit. on pp. 3, 4).

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*. 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423 (cit. on pp. 3, 4).

[3] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. 2020, pp. 9459–9474. URL: https://arxiv.org/abs/2005.11401 (cit. on p. 2).

[4] J. M. Lourenço. *The `aidisclose` package: Generative AI disclosure checklist and statements*. Version 1.6.2. 2025. URL: https://ctan.org/pkg/aidisclose (cit. on p. iii).

[5] T. Mikolov, K. Chen, G. Corrado, and J. Dean. "Efficient Estimation of Word Representations in Vector Space". In: *1st International Conference on Learning Representations (ICLR)*. 2013. DOI: 10.48550/arXiv.1301.3781 (cit. on pp. 2, 3).

[6] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al. "Training Language Models to Follow Instructions with Human Feedback". In: *arXiv preprint arXiv:2203.02155* (2022). URL: https://arxiv.org/abs/2203.02155 (cit. on pp. 4, 5).

[7] Y. Suchikova, N. Tsybuliak, J. A. T. da Silva, and S. Nazarovets. "GAIDeT (Generative AI Delegation Taxonomy): A Taxonomy for Humans to Delegate Tasks to Generative Artificial Intelligence in Scientific Research and Publishing". In: *Accountability in Research* (2025), pp. 1–27. DOI: 10.1080/08989621.2025.2544331 (cit. on p. iii).

[8] The LaTeX Project team. *LaTeX - A document preparation system*. 1985. URL: https://www.latex-project.org (visited on 2025-12-29) (cit. on p. iii).

[9] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, et al. "Llama 2: Open Foundation and Fine-Tuned Chat Models". In: *arXiv preprint arXiv:2307.09288* (2023). URL: https://arxiv.org/abs/2307.09288 (cit. on p. 5).

[10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. "Attention Is All You Need". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 30. 2017. DOI: 10.48550/arXiv.1706.03762 (cit. on pp. 3, 4).

A

Appendix Title

B

Appendix Title